1.3 Special Relativity

Special relativity is a theory of transformations between inertial reference frames, proposed by Albert Einstein in 1905. Through a deductive approach, Einstein arrived at the same transformation equations as those obtained by Lorentz. Since special relativity will be used extensively in the subsequent discussion of local field theory, we provide a brief derivation here.

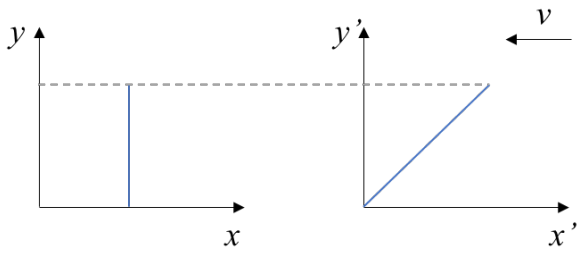


Fig.1.3. Photon trajectories in two reference frames

As illustrated in Fig.1.3, consider the trajectories of a photon as observed from two reference frames in relative motion. Let the relative velocity between the two frames be v. From the perspective of frame S', the photon travels a distance $\Delta x' = v\Delta t'$ along the x'-direction. According to the invariance of the speed of light in inertial frames, we have

$$\sqrt{x'^2 + y'^2} = \sqrt{x'^2 + y^2} = \sqrt{v^2\Delta t'^2 + c^2\Delta t^2} = c\Delta t' \tag{1.3.1}$$

From this equation, we obtain the relationship between the time intervals measured in the two reference frames:

$$\Delta t' = \frac{\Delta t}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} \tag{1.3.2}$$

We now change the photon's direction of propagation so that it travels along the x-axis. In this case, we obtain

$$\begin{cases} \Delta x = c\Delta t \\ \Delta x' = c\Delta t' \end{cases} \tag{1.3.3}$$

Substituting Eq. (1.3.2) into Eq. (1.3.3), we obtain

$$\Delta x' = \frac{\Delta x}{\sqrt{1-\left(\frac{v}{c}\right)^2}} \tag{1.3.4}$$

Furthermore, since reference frame S' moves relative to frame S with velocity v, an arbitrary spatial coordinate x' in frame S' can be expressed as

$$x' = \frac{x - vt}{\sqrt{1-\left(\frac{v}{c}\right)^2}} \tag{1.3.5}$$

By the principle of relativity, the inverse transformation can be written as

$$x = \frac{x' + vt'}{\sqrt{1-\left(\frac{v}{c}\right)^2}} \tag{1.3.6}$$

Substituting Eq. (1.3.5) into Eq. (1.3.6), we obtain the relationship between the time coordinates:

$$t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1-\left(\frac{v}{c}\right)^2}} \tag{1.3.7}$$

Combining Eqs. (1.3.5) and (1.3.7), we arrive at

$$x' = \frac{x - vt}{\sqrt{1-\left(\frac{v}{c}\right)^2}}, \qquad t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1-\left(\frac{v}{c}\right)^2}} \tag{1.3.8}$$

Equation (1.3.8) is the **Lorentz transformation**. Examining this transformation, we find that

$$c^2t'^2 - x'^2 = \frac{c^2t^2 - 2vxt + \frac{v^2x^2}{c^2} - x^2 + 2vxt - v^2t^2}{1-\left(\frac{v}{c}\right)^2} = c^2t^2 - x^2 \tag{1.3.9}$$

This equation shows that the quantity $c^2t^2 - x^2$ remains unchanged under a transformation between reference frames. A quantity that is invariant under such transformations is called a **scalar**. We now introduce a four-dimensional vector, $x^i = (ct, x, y, z)$ and define the metric to be the Minkowski metric:

$$\eta_{ij} = \begin{cases} 1 & i = j = 0 \\ -1 & i = j \neq 0 \\ 0 & i = j = 1,2,3 \end{cases} \tag{1.3.10}$$

Using Eq. (1.3.10) and considering the infinitesimal form, the invariant quantity in Eq. (1.3.9) can be expressed as

$$\eta_{ij}dx^i dx^j \tag{1.3.11}$$

A closer examination of the Lorentz transformation reveals that it is precisely a transformation that preserves the scalar in Eq. (1.3.11). In other words, it is a **length-preserving transformation**. Geometrically, this transformation can be understood in terms of a rotation of the reference frame in spacetime.

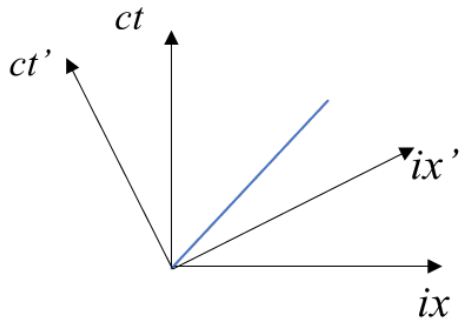


Fig.1.4. Transformation between reference frames

As the relative velocity between the two reference frames increases, the angle between their axes in Fig.1.4 also increases. When the ct'-axis coincides with the worldline of the particle, the frame S' becomes the particle's local inertial frame. In this frame, the particle is at rest, and only time elapses. If the particle carries a clock, the time interval recorded by that clock corresponds to the length of the event's worldline in spacetime. This time interval is called the **proper time**, denoted by τ. Substituting this into Eq. (1.3.11), we obtain

$$ds^2 = c^2 d\tau^2 = \eta_{ij}dx^i dx^j = c^2 dt^2 - dx^2 \tag{1.3.12}$$

As discussed in Section 1.2, both the Lagrangian and the action are scalars. Consequently, the integration measure in the relativistic action should also be expressed in terms of a scalar quantity. According to Eq. (1.3.12), the invariant quantity in special relativity is the spacetime interval along the worldline. Therefore, the action for a particle in special relativity should take the form

$$S = \int \alpha ds \tag{1.3.13}$$

Substituting Eq. (1.3.12) into Eq. (1.3.13), we obtain

$$S = \int \alpha c \sqrt{1 - \left(\frac{v}{c}\right)^2}\, dt \tag{1.3.14}$$

The corresponding relativistic Lagrangian is therefore

$$L = \alpha c \sqrt{1 - \left(\frac{v}{c}\right)^2} \tag{1.3.15}$$

In the low-velocity limit, this becomes

$$L = \alpha c - \frac{\alpha}{2c} v^2 \tag{1.3.16}$$

According to Eq. (1.2.3), an additive constant in the Lagrangian does not affect the equations of motion. We may therefore discard the first term in Eq. (1.3.16). To recover the classical expression in Eq. (1.2.5) in the low-velocity limit, the constant (\alpha) must satisfy

$$\alpha = -mc \tag{1.3.17}$$

Consequently, Eq. (1.3.15) becomes

$$L = -mc^2 \sqrt{1 - \left(\frac{v}{c}\right)^2} \tag{1.3.18}$$

Using Eqs. (1.2.7) and (1.3.18), we obtain the energy of a freely moving particle in special relativity:

$$\varepsilon = \frac{\partial \left[-mc^2 \sqrt{1 - \left(\frac{v}{c}\right)^2}\right]}{\partial v} \cdot v - \left[-mc^2 \sqrt{1 - \left(\frac{v}{c}\right)^2}\right] = \frac{mc^2}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} \tag{1.3.19}$$

The corresponding momentum is

$$\boldsymbol{p} = \frac{\partial \left[-mc^2 \sqrt{1 - \left(\frac{v}{c}\right)^2}\right]}{\partial \boldsymbol{v}} = \frac{m\boldsymbol{v}}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} \tag{1.3.20}$$

Combining Eqs. (1.3.19) and (1.3.20), we obtain an expression for the velocity in terms of energy and momentum:

$$\boldsymbol{v} = \frac{\boldsymbol{p}c^2}{\varepsilon} \tag{1.3.21}$$